

Take Home Challenge – Lead Mobile Developer

iOS Focused Challenge

Submission Deadline: 3 business days

You should supply the code for review and be ready to reference for the follow-up technical interview and be ready to generate the same solution for different platforms like Android natively or KMP. The purpose of this assignment is to provide us a sample of your code and architecture approach. It should be representative of what you would do as a professional level. It should be representative of what you would do as a professional during your day to day on the job. Please take the time to review the grading criteria below.

How will I be graded?

Architecture & Design

- Implementation of **MVVM pattern** in a **Modular Architecture**
- Application of **SOLID principles**
- Usage of **dependency injection**

Code Quality

- **No code duplication**
- **Good naming conventions One class/object per file**
- Avoid **hardcoded values**

Networking

- Implements a proper **networking layer**
- Demonstrate **error handling** with actual user-facing feedback

Project Structure & Organization

- **Well-organized folder structure**
- Uses **custom views** and reuse them accross the project

Testing

- Use **mocks**, ideally in separate dedicated mock files (not crammed into the test file itself)

Documentation

- Code includes **inline documentation**

We incentivize you to perform mindful choices in your software solution and document them in a README, so the evaluators can follow your train of thought. If you cannot meet all the requirements (due to time constraints or any other reasons), be sure to implement that as well. If you provide a list of things you would pick to implement next if you were to maintain this project for a long-time, that will be taken as a plus.

Use of AI

We strive to be at the forefront of AI use in our projects, which is why we allow the use of AI in the development of this test. However, we would like to know how you used it in a project like this, so we ask that you include an AI section in the README file where you describe which tools you used and how you utilized AI to help you develop this project.

Take Home Assignment - Local Sake Shops (iOS Native)

For this assignment, you will build a simple app that allows the user to see

- 1) A screen with a list of local sake shops and
- 2) A details screen for a specific sake shop.

Requirements:

The data is going to be supplied in JSON format.

The list page should contain items with sake shop names, address, and star ratings.

Whenever a sake shop from the list is tapped, the user should be directed to the details page of that shop. The details page should contain details about the sake shops, including:

- Shop name.
- Shop picture.
- A brief description.
- A rating in stars.
- Address (clickable, should open address in a Maps app or browser).
- A button or link to visit the shop's website in default web browser or in a custom tab.

Instructions:

1. Build a list page component that displays all items from the supplied JSON data you got together with this assignment.
2. You will be evaluated based on at least some of the following:
 - a. Modular Architecture Approach
 - b. Clean Architecture
 - c. SwiftUI must be used as the native development framework for this assignment.
 - i. **UIKit submission will not be accepted.**
 - d. Re-usability of your solution (could someone re-implement your solution easily for a different type of data / different API?).
 - e. Good UX and UI will be considered pluses, but you don't need to spend a very long time pixel-perfecting the look and feel (we don't expect you to be a designer). Follow material design guidelines and you're good.
 - f. Other factors include: any tests included, ui / snapshot tests, component structure, standards, performance and stability, etc.
3. Feel free to show off your coding ability by adding any level of appropriate complexity.
4. Feel free to use any AI tools for the development of this project.
5. Include a README that details information about your project, dependencies, setup, AI use and requirements.

Submission

Your app should be able to run in simulators. Please zip the project folder and include a simple README on setting up dependencies if your solution requires it.

Sake Shop

Response:

[{ "name": "信州スシサカバ 寿しなの", "description": "Sushi bar with a variety of sake options.", "picture": "[http://ts1.mm.bing.net/th?id=OIP.GURnZicaENMLYBMZN9k1LwHaFS&pid=15.1](\"http://ts1.mm.bing.net/th?id=OIP.GURnZicaENMLYBMZN9k1LwHaFS&pid=15.1\")", "rating": 4.0, "address": "〒380-0824 長野県長野市南長野南石堂町 1421", "coordinates": [36.644257, 138.18668], "google_maps_link": "[https://maps.app.goo.gl/4fYMDSfNd6ocsDwt6](\"https://maps.app.goo.gl/4fYMDSfNd6ocsDwt6\")", "website": "[https://www.sushinano.com/](\"https://www.sushinano.com/\")" }, { "name": "長野県酒類販売(株)", "description": "Wholesale sake distributor.", "picture": "[https://www.nagano-sake.com/common/images/front/drinks-xxl@2x.jpg](\"https://www.nagano-sake.com/common/images/front/drinks-xxl@2x.jpg\")", "rating": 4.2, "address": "〒380-0835 長野県長野市新田町 1464", "coordinates": [36.629883, 138.21141], "google_maps_link": "[https://maps.app.goo.gl/wRD6LRQc7Ct9QXMG8](\"https://maps.app.goo.gl/wRD6LRQc7Ct9QXMG8\")", "website": "[http://www.nagano-sake.com/](\"http://www.nagano-sake.com/\")" }, { "name": "Kadode", "description": "Japanese restaurant with high-quality sake.", "picture": "[https://rimage.hitosara.com/gg/image/0004030762/0004030762I1_1024x767y.webp](\"https://rimage.hitosara.com/gg/image/0004030762/0004030762I1_1024x767y.webp\")", "rating": 4.4, "address": "〒380-0824 長野県長野市南長野南石堂町 1421", "coordinates": [36.648273, 138.18724], "google_maps_link": "[https://maps.app.goo.gl/4z71FzXNYYM5oT2GA](\"https://maps.app.goo.gl/4z71FzXNYYM5oT2GA\")", "website": "[https://hitosara.com/0004030762/](\"https://hitosara.com/0004030762/\")" }, { "name": "Midori Nagano", "description": "Shopping center with a variety of sake shops.", "picture": null, "rating": 4.0, "address": "〒380-0824 長野県長野市南長野南石堂町 1421", "coordinates": [36.64396, 138.18893], "google_maps_link": "[https://maps.app.goo.gl/yPqYieSzdpsyCSd98](\"https://maps.app.goo.gl/yPqYieSzdpsyCSd98\")", "website": "[https://www.eki-midori.com/nagano/](\"https://www.eki-midori.com/nagano/\")" }, { "name": "Endo Brewery", "description": "Historic brewery known for its Keiryu brand sake.", "picture": "[https://www.keiryu.jp/img_201904/head_parts/shop_img.png](\"https://www.keiryu.jp/img_201904/head_parts/shop_img.png\")", "rating": 4.5, "address": "〒382-0086 長野県須坂市大字須坂 29", "coordinates": [36.648273, 138.18724], "google_maps_link": "[https://maps.app.goo.gl/f288RPXsgHRch3297](\"https://maps.app.goo.gl/f288RPXsgHRch3297\")", "website": "[https://www.keiryu.jp/](\"https://www.keiryu.jp/\")" }, { "name": "Masuichi Brewery", "description": "Brewery located in Obuse, known for its traditional sake.", "picture": "[https://gigaplus.makeshop.jp/masuichi/assets_v2/img/common/slide1.jpg](\"https://gigaplus.makeshop.jp/masuichi/assets_v2/img/common/slide1.jpg\")", "rating": 4.3, "address": "〒381-0201 長野県上高井郡小布施町大字小布施 807", "coordinates": [36.648273, 138.18724], "google_maps_link": "[https://maps.app.goo.gl/qrZekaJq8qP3yMiy7](\"https://maps.app.goo.gl/qrZekaJq8qP3yMiy7\")", "website": "[https://www.masuichi.com/](\"https://www.masuichi.com/\")" }, { "name": "Miyasaka Brewery", "description": "Brewery in Suwa City, offering a variety of sake.", "picture": "[https://www.masumi.co.jp/en/wp/wp-content/uploads/2025/02/top_1.jpg](\"https://www.masumi.co.jp/en/wp/wp-content/uploads/2025/02/top_1.jpg\")", "rating": 4.2, "address": "〒392-0004 長野県諏訪市諏訪 2 丁目 9-25", "coordinates": [36.648273, 138.18724], "google_maps_link": "[https://maps.app.goo.gl/wJTiquoJ4Qj8xsUp6](\"https://maps.app.goo.gl/wJTiquoJ4Qj8xsUp6\")", "website": "[http://www.masumi.co.jp/](\"http://www.masumi.co.jp/\")" },

```
{ "name": "Tamamura Honten", "description": "Brewery in Yamanouchi, known for its sake and beer.", "picture": "https://img07.shop-pro.jp/PA01336/957/slideshow/slideshow_img1.jpg?cmsp_timestamp=20160118182621", "rating": 4.6, "address": "〒381-0401 長野県下高井郡山ノ内町大字平穏 1163", "coordinates": [36.648273, 138.18724], "google_maps_link": "https://maps.app.goo.gl/htbv7LKDc9N3aHZ9A", "website": "http://www.tamamura-honten.co.jp/" }, { "name": "Shinshu Kuraudo", "description": "Specialty shop offering sake from around Nagano.", "picture": "https://www.nagano-wine.jp/store/.assets/DSC6668.jpg", "rating": 4.1, "address": "〒380-0824 長野県長野市南長野南石堂町 1421", "coordinates": [36.648273, 138.18724], "google_maps_link": "https://maps.app.goo.gl/SuUjW4R3wzuj3YVWA", "website": "https://www.nagano-wine.jp/store/pickup/post-34/" }, { "name": "Hokuan Brewery", "description": "Small brewery in Omachi City, known for its Daikokumasamune sake.", "picture": "https://hokuan.co.jp/wp/wp-content/themes/hokuan/img/bg01.jpg", "rating": 4.0, "address": "〒398-0002 長野県大町市大町 2340-1", "coordinates": [36.648273, 138.18724], "google_maps_link": "https://maps.app.goo.gl/GQrMNV06YoVEvCPU6", "website": "https://hokuan.co.jp/" } ]
```